# Testing Silicon Sampling in Public Opinion Research: Cross-National and Sociodemographic Patterns in LLM-Generated Survey Responses in the United States, Argentina, and Uruguay

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## Abstract

## 1. Introduction

Public opinion surveys have long been a foundational tool for studying political attitudes, social values, and behavioral patterns. In recent years, however, traditional survey research has faced growing challenges, including declining response rates, rising costs, and difficulties in reaching specific subpopulations (Bisbee et al., 2024; Boelaert et al., 2025). In this context, the rapid development of large language models (LLMs) such as GPT-3 and GPT-4 has sparked considerable interest in their potential to complement or partially substitute traditional data collection. The capacity of these models to generate fluent, human-like text has prompted researchers to ask whether they can also approximate the structure of human opinion when prompted to adopt specific demographic profiles.

This emerging practice, commonly referred to as silicon sampling, involves conditioning LLMs on sociodemographic attributes and prompting them to respond to survey items as if they were members of the corresponding population (Argyle et al., 2022). Early studies have reported encouraging levels of correspondence between synthetic and human responses in specific contexts, particularly in the United States (Argyle et al., 2022; Sun et al., 2024). At the same time, a growing body of critical work has documented important limitations in the stability, representativeness, and reproducibility of LLM-generated survey data (Bisbee et al., 2024; Boelaert et al., 2025), as well as systematic cultural and demographic biases in model outputs (Tao et al., 2024; Qu and Wang, 2024). The empirical evidence to date is therefore mixed, and the conditions under which silicon sampling produces reliable results remain unclear.

A notable feature of this literature is its strong concentration on English-speaking and predominantly U.S. contexts. Comparatively little is known about how LLMs perform when asked to simulate respondents in other national settings, and even less about whether they reproduce not only aggregate response distributions but also the internal sociodemographic structure that characterizes public opinion in those contexts. This gap is particularly relevant for Latin America, a region that has received limited attention in silicon sampling research despite its distinctive political trajectories and public opinion dynamics.

This paper addresses these issues by evaluating silicon sampling in the United States, Argentina, and Uruguay, using political interest as a substantive case. Drawing on data from Wave 7 of the World Values Survey, we generate synthetic responses to the political interest item across ten language models that vary in architecture, parameter size, and training regime, and compare them with empirical survey responses along three analytical dimensions: aggregate correspondence, cross-national variation, and internal sociodemographic structure. Our results show that while several models approximate aggregate distributions reasonably well, they tend to compress response variability, fail to consistently reproduce sociodemographic gradients, and exhibit larger distortions in the two Latin American contexts than in the United States. These findings contribute to ongoing debates about the validity, portability, and limits of silicon sampling for cross-national public opinion research.

## 2. Definition and Mechanisms of Silicon Sampling

Silicon Sampling refers to the practice of generating survey responses from LLMs by conditioning them on specified demographic or psychographic profiles. These profiles may include age, gender, ethnicity, geographic location, political affiliation, or other relevant attributes. The goal is to approximate how individuals with those traits might respond to survey items.

Sampling strategies vary in structure and intent. Random silicon sampling (RSS) involves drawing respondent profiles from real-world demographic distributions, such as national census data, and using those profiles to prompt the model. Stratified or quota-based approaches ensure that synthetic samples include fixed proportions of key subgroups, mirroring common survey design practices. These strategies aim to improve representativeness within synthetic samples, although they rely on assumptions about how prompts influence LLM outputs.

Other implementations adopt a purposive design, creating personas to examine specific subpopulations or scenarios rather than statistical representativeness. This approach is often used to explore how particular categories of individuals—such as undecided voters or patients with chronic conditions—might respond to different questions or messages. In some cases, researchers employ iterative sampling and filtering techniques, where multiple outputs are generated for a single profile, and only those meeting specific criteria are retained.

The process typically involves prompting the model to "act as" a respondent with given characteristics (e.g., "a 35-year-old Latino male living in Texas who voted for Biden"). These prompts are crafted to encourage consistent and contextually appropriate responses. Some studies use fixed persona libraries across survey items, while others dynamically adjust prompts or personas as the study progresses. Hybrid designs, which combine synthetic and human data, have also been introduced, allowing researchers to compare patterns across data types or use synthetic samples to pretest instruments. While these strategies aim to improve representativeness, recent work has documented important limitations in the stability and validity of LLM-generated survey responses. Bisbee et al. (2024) show that synthetic responses produced by ChatGPT are highly sensitive to minor prompt variations and exhibit substantial drift over time, raising concerns about reproducibility. Boelaert et al. (2025) further document a tendency toward response homogenization, in which LLM-generated samples display considerably less variance than comparable human samples and tend to underrepresent minority positions. Lin (2025) systematizes a series of methodological fallacies involved in substituting LLMs for human participants, cautioning against treating synthetic outputs as direct equivalents of survey data.

## 3. Applications of Silicon Sampling

Silicon Sampling has been applied in several areas of social science research. In political science, studies have evaluated whether LLM-generated responses can approximate patterns found in traditional survey data. For instance, Argyle et al. (2022) used GPT-3 to generate responses to the American National Election Survey (ANES) and compared them with observed survey results. Their analysis focused on alignment at both aggregate and subgroup levels. Sun et al. (2024) used random silicon sampling to replicate distributions from the 2020 U.S. presidential election, reporting that model outputs were broadly similar to public opinion data along several demographic dimensions. Geng et al. (2024) explored the influence of prompt design in shaping synthetic responses, highlighting that outcomes varied significantly with seemingly minor changes in language or structure.

In policy research, synthetic samples have been used to simulate responses to policy proposals such as Universal Basic Income, climate regulation, and immigration reform. These studies typically use controlled prompts to test the effects of different framings or contextual cues on support or opposition. In some cases, researchers explore within-group variation by generating multiple responses from personas with similar demographic characteristics but slightly altered prompts.

In the domain of consumer behavior and marketing, synthetic personas have been used to explore product preferences, brand perception, and response to advertising content. Sarstedt et al. (2024) investigated the plausibility of LLM-generated consumer feedback, finding some correspondence with known patterns in

human response data. These applications often serve as exploratory tools in early-stage product testing or messaging development.

In educational settings, synthetic respondents have been used to model hypothetical student feedback on instructional materials, simulate responses to curriculum changes, and examine engagement across different learner profiles. This approach is used primarily for exploratory purposes or to assess system responsiveness in automated tutoring environments.

Public health studies have applied silicon sampling to investigate responses to messaging around vaccination, mental health stigma, and health service delivery. For example, synthetic cohorts have been used to test reactions to vaccine information campaigns or alternative phrasing in mental health questionnaires. These studies often focus on content framing and comprehension.

Ethics and governance research has used synthetic personas in deliberative simulations and stakeholder exercises. These studies aim to understand how different kinds of individuals might reason through complex issues such as AI regulation or biomedical innovation. LLMs are prompted to simulate contributions in deliberative contexts, and the resulting outputs are analyzed thematically.

Qualitative research has employed LLM-generated text to simulate interview responses, stakeholder dialogues, and ethnographic vignettes. Amirova et al. (2024) compared LLM-generated interviews with transcripts from real interviews in UK health services. While some thematic alignment was noted, differences in tone, spontaneity, and the depth of narrative structure were also observed. These findings highlight both the potential utility and limitations of synthetic data in qualitative analysis.

A parallel line of research has examined cultural and demographic biases in LLM outputs that are particularly relevant for the use of these models in cross-national survey simulation. Tao et al. (2024) and Atari et al. (2023) show that LLMs tend to express values aligned with WEIRD societies, and particularly with English-speaking and Protestant European countries, with greater divergence observed for non-WEIRD populations. Closely related to the present study, Qu and Wang (2024) use the World Values Survey to evaluate ChatGPT's performance in simulating public opinion across countries, and find that the model reproduces empirical patterns more accurately in Western, English-speaking, and developed contexts—particularly the United States—than elsewhere, with additional disparities across demographic groups defined by gender, ethnicity, age, education, and social class. A complementary strand of work has documented that LLMs reproduce and at times amplify sociodemographic stereotypes when conditioned on personas. Cheng et al. (2023) show that persona-based generation often yields stereotyped portrayals, while Kotek et al. (2023) document systematic gender stereotyping in occupational associations.

Despite this growing literature, most existing evaluations have focused on English-speaking contexts and, in particular, on the United States. Less is known about whether LLM-generated survey responses can reproduce not only aggregate distributions but also cross-national differences and the internal sociodemographic structure of public opinion in non-WEIRD settings. This paper addresses that gap by comparing synthetic and survey-based responses on political interest in the United States, Argentina, and Uruguay, examining both aggregate correspondence and the extent to which models reproduce the social patterning of political engagement across these three national contexts.

## 4. Data, Design, and Analytical Strategy

### 4.1 Data source and substantive focus

We evaluate silicon sampling using data from the World Values Survey (WVS), Wave 7, a large cross-national survey that collects comparable data on values, attitudes, and beliefs across a wide range of countries. The WVS is widely used in comparative research because of its standardized questionnaires, extensive geographic coverage, and well-documented sampling procedures, which make it a valuable benchmark for assessing patterns of public opinion across different social and national contexts. In addition to providing the empirical reference for comparison, the survey also supplies the sociodemographic variables used to construct the synthetic respondent profiles.

Our analysis focuses on Q199, an item measuring respondents' interest in politics. The question asks respondents how interested they are in politics and offers a four-point response scale ranging from "Not at all interested" to "Very interested." Political interest is a central concept in the study of public opinion and political behavior, often interpreted as an indicator of political engagement and attentiveness to public affairs. It has been extensively studied in the comparative literature and is known to vary both across countries and across social groups within countries.

This variable is particularly suitable for evaluating silicon sampling for several reasons. First, it represents a well-established attitudinal dimension, making it possible to compare synthetic outputs with a large body of existing empirical research. Second, the distribution of political interest typically exhibits substantive variation across countries, reflecting differences in political institutions, media environments, and levels of civic engagement. Third, political interest is also strongly associated with sociodemographic characteristics such as education, age, occupation, and social class, allowing us to assess not only whether language models approximate overall response distributions, but also whether they reproduce the internal social structure of public opinion.

To examine these issues, we focus on three national contexts: the United States, Argentina, and Uruguay. These countries were selected in order to capture variation in political and institutional settings while maintaining comparability in survey design and question wording. The United States represents a highly studied case in survey research and provides a useful reference point for evaluating synthetic responses. Argentina and Uruguay, in turn, offer two Latin American contexts with distinct political trajectories and public opinion dynamics. By comparing these three cases, we assess whether LLM-generated responses reproduce both cross-national differences in political interest and within-country patterns of sociodemographic differentiation.

### 4.2 Synthetic respondent design

The unit of analysis in the synthetic data generation process is a synthetic respondent profile, defined by a combination of sociodemographic attributes embedded in the prompt provided to the language model. Each synthetic respondent is represented as a profile with specific demographic and social characteristics.

Synthetic profiles were constructed using a set of core attributes derived from the WVS: country, age, sex, education, occupation, and subjective social class. Each profile was assigned to one of the three national contexts examined in this study—the United States, Argentina, or Uruguay—and combined with values for the remaining sociodemographic variables. These attributes were chosen because they represent commonly used dimensions of social differentiation in survey research and are frequently associated with variation in political attitudes and engagement.

To guide the selection of attributes included in the synthetic respondent profiles, we conducted an exploratory analysis using the WVS dataset. Specifically, we estimated a Random Forest model to examine the relative importance of several sociodemographic variables in explaining variation in political attitudes, including interest in politics (Q199), evaluations of different political systems (Q235–Q239), and left–right ideological self-placement (Q240). Variable importance measures were used to identify which attributes contributed most strongly to explaining variation in these outcomes. The resulting importance plots (see Appendix Figure XX) indicate that country, age, education, occupation, and subjective class consistently appear among the most influential predictors, while other variables display weaker or less consistent contributions across outcomes.

Based on these exploratory results, we selected a parsimonious set of sociodemographic attributes—country, age, sex, education, occupation, and subjective class—to define the synthetic respondent profiles used in the prompting procedure. This choice allowed us to retain variables that are empirically associated with political attitudes while keeping prompts sufficiently concise to avoid excessive contextual complexity.

Each synthetic respondent profile was embedded into a structured prompt describing the individual's demographic and social characteristics. The prompt instructed the language model to answer the survey question as if it were a person with the specified attributes and answer the survey question from the perspective of a person with those attributes. By systematically varying these attributes across profiles, the design enables us

to evaluate whether LLM-generated responses reproduce not only aggregate distributions but also structured variation across sociodemographic groups.

### 4.3 Models and prompting procedure

Synthetic responses were generated using several large language models (LLMs) that differ in architecture, training data, and parameter size. Including models with different characteristics allows us to evaluate whether patterns observed in silicon sampling are consistent across model families or whether they vary depending on model scale and design.

Table XX summarizes the models used in the analysis together with their approximate parameter size and the size categories employed.

Table 1: Large language models used in the analysis, model type, approximate parameter size, and size category.

| Model | Type | Model size | Category |
|---|---|---|---|
| DeepSeek-R1 | Open-weight | 1.5B | Small |
| Qwen 3 | Open-weight | 1.7B | Small |
| llama3.2 | Open-weight | 3B | Small |
| gemma3 | Open-weight | 4B | Small |
| Llama 3.1 | Open-weight | 8B | Small |
| GPT-4.1 Mini | Proprietary | ~15B | Medium |
| Gemini 2.5 Flash | Proprietary | ~25B | Medium |
| gemma3 | Open-weight | 27B | Medium |
| Qwen 3 | Open-weight | 30B | Medium |
| GPT-4.1 | Proprietary | ~175B | Large |

The models included represent both open-weight models and proprietary systems, covering a wide range of parameter sizes. To facilitate comparison, models were grouped into three broad categories according to their scale: Small (up to 10B parameters), Medium (10 to 30B parameters), and Large (more than 30B parameters). This categorization allows us to explore whether the ability of LLMs to reproduce survey response patterns varies with model capacity.

In general terms, larger models are expected to capture more complex statistical regularities from their training data, which could potentially improve their ability to approximate patterns observed in real survey responses. However, the relationship between model size and the sociological plausibility of generated responses remains an open empirical question. By comparing models across these size categories, the analysis allows us to examine whether increases in model scale are associated with improvements in the reproduction of aggregate distributions and sociodemographic patterns.

For each synthetic respondent profile, the model was prompted to answer the survey question while adopting the perspective of the specified individual. The prompt included a short description of the respondent's demographic and social characteristics (country, age, sex, education, occupation, and subjective social class) and instructed the model to answer the survey question as if it were a person with those attributes.

The prompt structure consisted of three elements. First, a brief description of the synthetic respondent profile was provided. Second, the survey question was presented using wording consistent with the original WVS item. Third, the set of valid response options was explicitly included in the prompt to constrain the model's output to the same response categories used in the survey.

Responses were generated independently for each model and respondent profile. By keeping the prompt structure constant across models, the design allows us to compare the resulting outputs while minimizing variation introduced by prompt design. The resulting dataset therefore consists of LLM-generated survey

responses associated with synthetic respondent profiles, which can be analyzed using the same procedures commonly applied to survey data.

This procedure produces a dataset of synthetic responses associated with defined respondent profiles. Because the structure of the prompts and response options mirrors the original survey item, the resulting outputs can be analyzed using the same descriptive and statistical techniques commonly applied to survey data. This enables a direct comparison between synthetic and observed responses at both the aggregate level and across sociodemographic subgroups.

### 4.4 Analytical strategy

To evaluate the performance of silicon sampling, we compare synthetic and observed responses along three analytical dimensions: aggregate correspondence, cross-national variation, and sociodemographic structure. Each dimension targets a distinct aspect of what it would mean for a language model to adequately approximate human survey data, ranging from the reproduction of marginal frequencies to the recovery of the internal social patterning of opinion.

First, we examine aggregate correspondence by comparing the overall distribution of synthetic responses with the empirical distribution observed in the WVS for each country. We summarize this correspondence using the Mean Absolute Error (MAE) between response proportions across the four categories of the political interest scale. To complement this measure and characterize the dispersion of model outputs, we also report the normalized entropy and the effective number of categories of each response distribution. The combination of these indicators allows us to distinguish between models that approximate the empirical distribution and those that concentrate responses in a single category, a phenomenon described in the recent literature as response homogenization or modal collapse (Boelaert et al., 2025; Dominguez-Olmedo et al., 2024).

Second, we examine cross-national variation by assessing whether models reproduce the systematic differences in political interest observed across the United States, Argentina, and Uruguay. This involves comparing both the central tendencies of synthetic responses across countries and the within-model variation in MAE and entropy across national contexts. This dimension addresses the extent to which models are responsive to the national context specified in the prompt, an aspect that prior work has linked to the predominantly Western and English-speaking composition of LLM training data (Tao et al., 2024; Qu and Wang, 2024).

Third, we examine sociodemographic structure by estimating separate logistic regression models predicting (dichotomized) political interest as a function of age, sex, education, and subjective social class. Models are estimated independently for the WVS and for each LLM, and the resulting coefficients are compared in terms of direction, magnitude, and dispersion across language models. Whereas aggregate measures capture whether models reproduce overall response frequencies, the regression-based comparison evaluates whether they also reproduce the conditional associations between sociodemographic attributes and political engagement that have been extensively documented in the comparative literature on political participation. This strategy follows the approach used in recent critical evaluations of silicon sampling (Bisbee et al., 2024), which have shown that even when aggregate distributions are reasonably approximated, regression-based associations may diverge substantially from those observed in human data.

Taken together, these three dimensions make it possible to distinguish between models that approximate aggregate response frequencies and those that also reproduce the relational and structural properties of empirical survey data.

## 5. Results

### 5.1 Aggregate response distributions and variability

We begin by examining the overall distribution of responses to the political interest question in the WVS and comparing these distributions with those generated by the language models.

A primary characteristic of the simulations performed is the systematic under representation of the extremes of the scales. This is particularly evident in the case of the category "Very interested" which appears as a meaningful proportion (>10%) in only one model (gemini-2.5-flash-Medium), with most models showing either complete absence or minimal presence (<5%) of this response option across all geographic contexts.

Figure 1: "Response distribution by model and country. Each bar shows the proportion of responses in five categories: 'Very interested' (dark teal), 'Somewhat interested' (light teal), 'Not very interested' (tan), 'Not at all interested' (dark brown), and invalid formats. Models are ordered by family and size. Note the extreme concentration in qwen3-Medium (ARG: 95% dark brown) versus balanced distribution in deepseek-r1-Small. The Uruguay advantage is evident in the higher proportion of light teal ('Somewhat interested') in the middle column compared to the left column for most models."

The negative extreme of this scale -"Not at all interested"- appears as the dominant response only in qwen3-Medium (where it indicates severe mode collapse), while "Not very interested" and "Somewhat interested" serve as the preferred modal categories for most models.

Three groups of models appear to be observable: models with balanced multi-category distributions (e.g., deepseek-r1-Small showing approximately 50-60% in the modal category across all countries), models with extreme concentration (e.g., qwen3-Medium displaying 95% "Not at all interested" responses in Argentina versus a more balanced 44% in the USA), and models with moderate concentration that remains stable across contexts (e.g., llama3.2-Small maintaining approximately 73-75% in "Not very interested" across all countries).

Uruguay, seems to have somekind of advantage: it shows consistently higher proportions of "Somewhat interested" responses in the middle column compared to Argentina, with seven of nine models showing increased color diversity in Uruguay.

Different variability metrics confirm the previous analysis, since it reveals heterogeneity in response variability across language models and countries. Model-specific behavioral patterns emerge clearly: gpt-4.1-mini-Medium exhibits extreme response concentration across all countries (normalized entropy 0.16-0.31, modal

percentages 81-93%), while deepseek-r1-Small maintains consistently high variability (entropy 0.62-0.67, effective categories 2.12-2.41).

However, significant geographic interactions complicate this picture. qwen3-Medium demonstrates dramatic cross-country behavioral shifts, operating near-deterministically in Argentina (entropy 0.14) while exhibiting the highest U.S. variability (entropy 0.66, effective categories 2.76). The Latin American samples display more pronounced bimodal distributions with clusters at behavioral extremes, whereas the U.S. sample shows smoother gradation across the variability spectrum.

These findings suggest that model response distributions reflect not merely architectural properties but complex interactions between training data composition, prompt interpretation mechanisms, and culturally-embedded survey content, raising fundamental questions about the portability of synthetic survey methodologies across demographic and linguistic boundaries.

### 5.2 Cross-national variation

To examine whether models reproduce systematic differences in political interest across national contexts, Figure XX reports the Mean Absolute Error (MAE) between the distribution of synthetic responses generated by each model and the empirical distribution observed in the WVS for each country. Lower values indicate closer aggregate correspondence with the data.

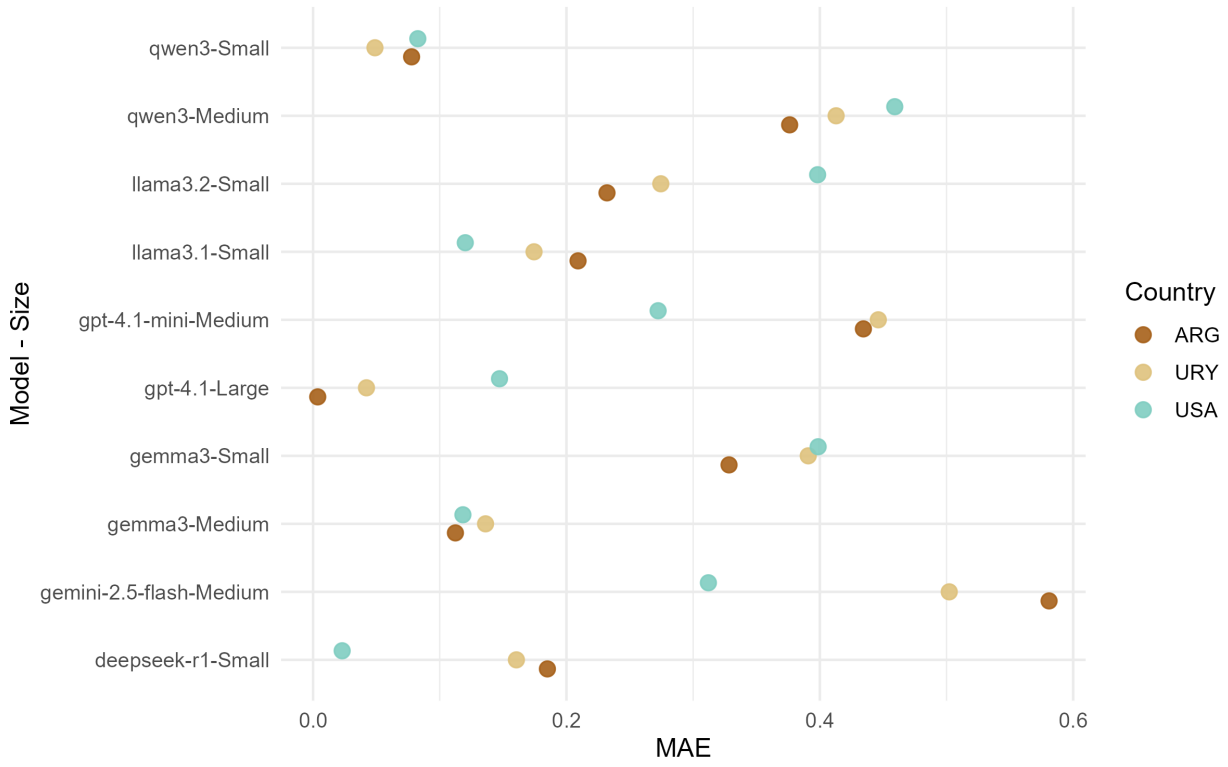


Figure 2: "Mean Absolute Error between the distribution of synthetic responses generated by each model and the empirical distribution observed in the WVS, by country. Each point represents the average absolute difference in response proportions across the survey categories. Lower values indicate closer correspondence between synthetic and observed distributions. Models are ordered by family and size, note the comparatively low error values for gpt-4.1-large and qwen3-small, and the substantially higher error levels for qwen3-Medium and gemma3-small across countries."

Among the best performing cases are qwen3-small in the United States, gpt-4.1-large in Argentina and

Uruguay, and qwen3-small in Uruguay, all of which exhibit relatively low MAE values, indicating a close approximation to the empirical response distributions. At the opposite extreme, qwen3-Medium shows consistently high error levels across all three countries, reflecting strong concentration of responses in a single category. Similarly, gemma3-Small displays comparatively large deviations from the empirical distribution in each national context. These patterns are consistent with the response concentration and modal collapse dynamics observed in the previous analysis.

Systematic differences across countries within the same model are also evident. In several cases, the same model produces noticeably different error levels across Argentina, Uruguay, and the United States, suggesting that aggregate fit depends not only on model architecture or parameter size but also on the interaction between the model and the national response distribution represented in the prompt.

### 5.3 Subgroup structure

To assess whether language models reproduce not only aggregate response distributions—as captured by the MAE—but also the social structure of opinion, we estimated separate logistic regression models for each source, that is, for the WVS and for each of the LLMs analyzed, using the same sociodemographic variables as predictors of political interest, operationalized as a dichotomous outcome. The regression framework allows us to move beyond average proportions and examine how the probability of the dependent variable—political interest—changes as each attribute varies, holding the others constant.

The effect of age in the WVS data is minimally positive across all three countries, meaning that older respondents tend to report slightly higher levels of political interest. LLMs generally reproduce the correct direction of this effect, though their coefficients approach zero, indicating that the distortion introduced is relatively modest.

In the case of education, the WVS shows a positive effect in all three countries: higher education is associated with greater political interest, a finding consistent with the broader literature on political participation. However, all the LLMs examined systematically overestimate this effect, in some cases substantially above the WVS baseline. While the direction is correct, the magnitude is consistently inflated. Considerable dispersion across models is also observed, particularly in Argentina.

Regarding sex, the WVS shows a consistent negative effect across all three countries, indicating that women report lower levels of political interest than men. LLMs perform considerably worse on this dimension: some models reverse the sign, others overestimate it, and confidence intervals are generally wide. A relevant geographic pattern also emerges—dispersion across models is greater in Argentina and Uruguay than in the United States, suggesting that models are less certain about how sex-based differences in political interest manifest in Latin American contexts.

With respect to social class, the WVS shows a small positive effect, with higher-class respondents reporting somewhat greater political interest, though the magnitude is modest. LLMs consistently overestimate this attribute, with the distortion being particularly pronounced in the United States. This implies an underlying representation of social class as a much stronger predictor of political interest than what the empirical data support.

In sum, the general pattern is that LLMs capture the direction of the sociodemographic gradients reasonably well but systematically exaggerate their magnitude, especially for education and social class. This reinforces the notion that models encode certain social stereotypes about who is interested in politics and how consequential their characteristics are. Sex is the most problematic predictor, as not even the direction is consistent across models. Moreover, distortions are more pronounced in the Latin American contexts than in the United States.

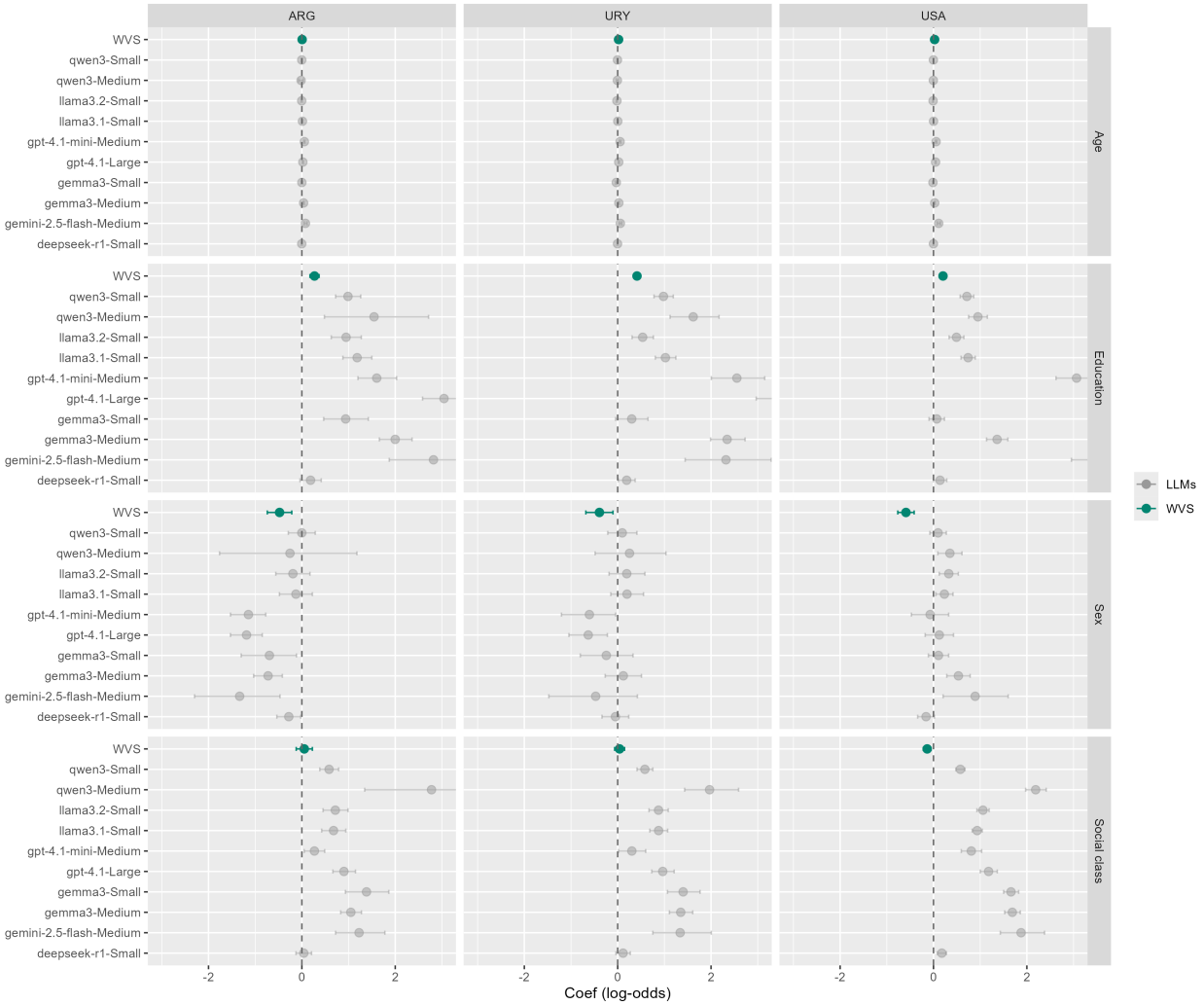


Figure 3: "Regression coefficients predicting political interest (dichotomous) by model and country. Each point represents the estimated coefficient for a given sociodemographic predictor, with horizontal lines indicating 95% confidence intervals. The green point corresponds to the empirical WVS baseline and the grey points correspond to each LLM. Models are ordered by family and size. Note the systematic overestimation of education and social class effects across all models, the near-zero age coefficients relative to the WVS baseline, and the inconsistent sign of the sex coefficient across all models considered. The greater dispersion of LLM coefficients in Argentina and Uruguay, compared to the United States, is particularly evident for the sex and social class variables."

# Conclusions

# Appendix

Table A1: Modal Response Concentration (%) by Model and Country

| Model and size | Modal cat. percent._ARG | Modal cat. percent._URY | Modal cat. percent._USA | Modal response_ARG | Modal response_URY | Modal response_USA |
|---|---|---|---|---|---|---|
| deepseek-r1-Small | 52.84 | 57.2 | 63.37 | Somewhat interested | Somewhat interested | Somewhat interested |
| gemini-2.5-flash-Medium | 76.87 | 75.0 | 73.54 | Somewhat interested | Somewhat interested | Somewhat interested |
| gemma3-Medium | 49.85 | 60.0 | 78.24 | Not very interested | Somewhat interested | Somewhat interested |
| gemma3-Small | 93.12 | 91.8 | 73.46 | Not very interested | Not very interested | Not very interested |
| gpt-4.1-Large | 60.42 | 49.3 | 65.33 | Not very interested | Not very interested | Somewhat interested |
| gpt-4.1-mini-Medium | 81.75 | 90.4 | 93.22 | Somewhat interested | Somewhat interested | Somewhat interested |
| llama3.1-Small | 77.57 | 68.2 | 53.70 | Not very interested | Not very interested | Somewhat interested |
| llama3.2-Small | 74.58 | 74.3 | 72.96 | Not very interested | Not very interested | Not very interested |
| qwen3-Medium | 94.92 | 80.0 | 44.18 | Not at all interested | Not at all interested | Not at all interested |
| qwen3-Small | 66.60 | 56.8 | 56.82 | Not very interested | Not very interested | Somewhat interested |

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